Fake Cafe Business Readme File

This is the introduction for the deliverable for Fake Cafe Business, specifically, the Data Hygiene Package.

This folder contains:
- The file you are reading right now
- The raw csv file that was given to me
- A cleaned version of the file
- An excel file that contains table version of the dataset
- An interactive Python Notebook (.ipynb) file that contains all of the code used to clean the data and an explanation of how it works
- A requirements file that lists all the possible requirements for your computer to properly run the coding file
- A matrix graph created during the cleaning process
- A bar graph created during the cleaning process
- A heatmap created during the cleaning process
- A data profiling report created during the cleaning process
- A Data Cleaning Audit Log that contains information about the entire cleaning process explained clearly for non-technical people

If you would like to see the completed work, open the excel File or cleaned csv file. Then read the Data Cleaning Audit Log for more information on the exact cleaning process.

If you are viewing this as a technical team member, open the excel File or cleaned csv file. Then open and skim through the .ipynb file. A full explanation on how to properly use and run the file is inside the file, but I have pasted the explanation at the bottom of this document just in case. You do not need to look through the requirements file, profiling report, or graphs/bars created, unless you are curious or need more information. You can then look through the Data Cleaning Audit Log for a more thorough explanation of the cleaning process.

As a reminder, you are able to freely copy, distribute, and utilize all the files provided to you. You may also edit the code in the Python file if you wish, though this is not recommended. You may wish to copy all the files in the original folder and then use the versions of the copied files for editing.

For more information or help, please feel free to contact me. Make sure to check the Data Cleaning Audit Log or .ipynb file, as your question may already be answered there.

Instructions for running .ipynb file

To properly run or view this file, you will need an application capable of working with Jupyter notebook files (.ipynb files). Below is a list of a few options you can use.

Option A: Anaconda Distribution
You can install a free, open source collection of data science coding tools called the Anaconda Distribution from their website: https://www.anaconda.com/download. This includes everything you need to view and run these files. This is the best option if you are planning to do more work with more datasets or are working on this particular file.

Option B: VS Code
If you prefer another editor, you can install Microsoft's Visual Studio Code along with its Python extensions. Go to https://code.visualstudio.com/download, then download the extensions using by clicking on the extensions button on the left sidebar, (the icon with the 4 squares) and searching up Jupyter Notebook. You will still need Python installed on your computer to run the code, which you can get by going to https://www.python.org/downloads/.

Option C: Google Colab
If you do not want to download software to your computer, you can upload the notebook file directly to Google Colab in your web browser https://colab.research.google.com/.

Once you have a proper application, we finally must make sure the project files and extra tools are ready.
Make sure that the original data file sent back to you, as listed in this file below, with the exact same name is in the same file or directory as this file that you're looking at. For example, this file was likely sent to you with a bundle of other files, including the original data file that was used here, all in a single folder. As long as this file and the raw data are still in the same folder, this .ipynb file should be able to find it and run properly. If that is not the case, you must find the original and but it back in this folder, or put them together in a new folder. There are other ways to solve this problem, but this is the simplest.

If you are getting a FileNotFound error or some other error that describes the file not being found, the files are either not together or the data file has had its name changed (which is making this file confused).

There are a couple of extra tools that this file needs to do its jobs, which are called libraries. The ones I always install and typically use are: pandas, numpy, plotly, openpyxl, matplotlib, seaborn, ydata-profiling, ipywidgets, and missingno.

Here's how to install them:

For the Anaconda Option
Open the Anaconda Navigator app from your computer applications menu. Click on Environments on the left panel, find the search box at the top, and type the name of the

library you need. Click the check box next to the library when it appears in the list, then click Apply at the bottom of the screen to install it. Repeat this process for each of the libraries that I listed above.

For the VS Code Option
Open VS Code on your computer and open a new Terminal from the top menu by clicking Terminal and then New Terminal. Type the line `pip install pandas numpy openpyxl matplotlib plotly seaborn ydata-profiling ipywidgets missingno` into the bottom window that pops up and press Enter. Wait a few moments while your computer downloads and installs all libraries at once.

For the Google Colab Option
Open your notebook file inside Google Colab using your web browser. Click the button to add a new Code cell at the very top of your file. Inside that new cell, type `!pip install pandas numpy openpyxl matplotlib plotly seaborn ydata-profiling ipywidgets missingno` including the exclamation point at the start. Press the play icon on the left side of the cell to run the line and install the libraries to your web workspace.

If you've done that and things still aren't working, check the requirements file.
A file named requirements.txt lists all the external Python libraries required to run the code without errors. You should install all the packages listed in that file to ensure everything works smoothly. Instead of installing libraries one by one, you can install every required package on the list at once by running this single command in your terminal or command prompt:

`pip install -r requirements.txt` (this requires python installed)

To check if a specific Python library is installed on your computer, use `pip list` or `pip show` in your command prompt:

Type `pip list` to view a complete list of every installed Python library and its version number.

Type `pip show` <library-name> (for example, pip show pandas) to check if a specific library is installed and view details about it.

For further assistance
If you are still stuck with a bug or can't get things working, try searching your exact problem or error message online. Here are some additional resources to help:
https://www.youtube.com/watch?v=WUeBzT43JyY
https://docs.jupyter.org/en/latest/install/notebook-classic.html